

1. a) How many 3-digit numbers can be formed from the digits 0, 1, 2, 3, 4, 5 and 6, if each digit can be only used once?

$$6 \cdot 6 \cdot 5 = 180$$

- b) How many of these are odd number?

$$3 \cdot 5 \cdot 5 = 75$$

- c) How many are greater than 330?

$$1 \cdot 3 \cdot 5 = 15$$

$$4 \cdot 6 \cdot 5 = 90$$

$$\underline{105}$$

2. A witness to a hit-and-run accident told the police that the license plate contained the letter RLH followed by 3 digits, the first of which is a 5. If the witness cannot recall the last 2 digits, but is certain that all 3 digits are different, find the maximum number of automobile registrations that the police may have to check.

$$\boxed{\text{RLH } 5 \underline{\quad} \underline{\quad} \underline{\quad}} = 72$$

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3. A random sample of 200 adults are classified below by sex and their level of education attained.

If a person is picked at random from this group, find the probability that

- a) the person is male, given that the person has a Secondary education

$$P(M|S) = \frac{28}{78}$$

	M	F	Total
E	38	45	83
S	28	50	78
C	22	17	39
Total	88	112	200

- b) the person does not have a college degree, given that the person is a female

$$P(E \cup S | F) = \frac{95}{112}$$

4. Before the distribution of certain statistical software every fourth CD is tested for accuracy. The testing process consist of running four independent programs and checking the result. The failure rate for the 4 testing program are respectively, 0.01, 0.03, 0.02, and 0.01
- $F_1 \quad F_2 \quad F_3$

- a) What was the probability that a CD was tested and failed any test?

$$P(F_1' \cap F_2' \cap F_3' \cap F_4') = 0.99 \cdot 0.97 \cdot 0.98 \cdot 0.99 = 0.93168 = P(\text{ALL PASS})$$

$$P(T \cap \text{ALL-PASS}') = 0.25 \cdot 0.06832 = 0.017079$$

- b) Given that a CD was tested, what is the probability that it failed program 2 or 3?

$$P(F_2 \cup F_3) = 0.03 + 0.02 - (0.03 \cdot 0.02) = 0.0494$$

- c) In a sample of 100, how many CDs would you expect to be rejected

$$100 \cdot P(T \cap \text{ALL-PASS}') = 100 \cdot 0.017079 = 1.7079 \approx 2$$

- d) Given a CD was defective, what is the probability that it was tested

$$P(T|D) = P(T) = 0.25$$

- 5 In a certain region of the country it is known from last experience that the probability of selecting an adult over 40 years of age with cancer is 0.05. If the probability of a doctor correctly diagnosing a person with cancer as having the disease is 0.78 and the probability of incorrectly diagnosing a person without cancer as having the disease is 0.06, what is the probability that a person is diagnosed as having cancer.

$$P(C) = 0.05 \quad P(C \cap D) = 0.78$$

$$P(C') = 0.95 \quad P(C' \cap D') = 0.06$$

$$(0.78 \cdot 0.05) + (0.06 \cdot 0.95) = \underline{\underline{0.096}}$$

- 6 Police plan to enforce speed limits by using radar trap at 4 different locations within the city limits. The radar traps at each of the locations L1, L2, L3, and L4 are operated 40%, 30%, 20%, and 30% of the time, and if a person who is speeding on his way to work has probabilities of 0.2, 0.1, 0.5, and 0.2, respectively, of passing through these locations, what is the probability that he will receive a speeding ticket?

$$(0.4 \cdot 0.2) + (0.3 \cdot 0.1) + (0.2 \cdot 0.5) + (0.3 \cdot 0.2) = \underline{\underline{0.27}}$$